Cell Tuition
1-1 PRIVATE TUTORS

STATISTICS

Time-Series Graphs

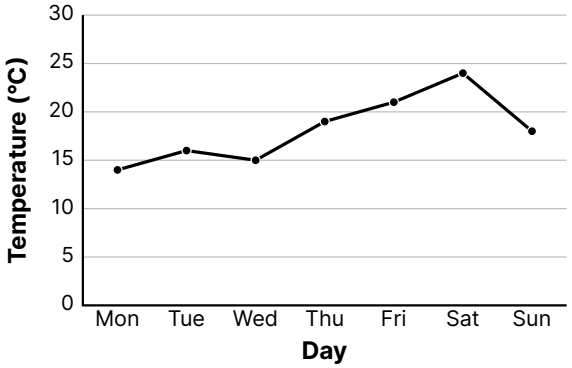
Name _____ **Date** _____ No calculator **Version 1** Scan to practise this topic online

KEY WORDS time-series graph *(kauwhata raupapa wā)* · increase *(pikinga)* · decrease *(hekenga)* · trend *(ia)*

A Reading a time-series graph ●○○

Remember: A time-series graph shows how something changes over time. Time goes along the bottom. Find the time, go up to the point, then read across to the scale.

1. The graph shows the temperature at midday in a town on each day of one week.



a) What was the temperature on Thu? _____

b) What was the temperature on Sun? _____

c) Which day was the warmest? _____

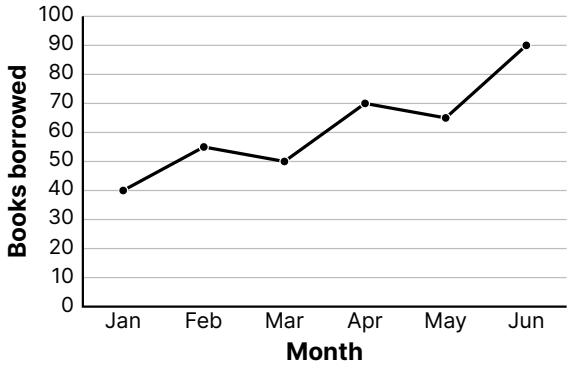
d) Which day was the coldest? _____

e) On which day was the temperature 16°C? _____

B Changes ●●○

Remember: When the line goes up, the amount is increasing. When it goes down, the amount is decreasing. A steeper line means a bigger change.

2. The graph shows the number of books borrowed from a school library each month.



a) How many more books were borrowed in Feb than in Jan?

b) Between which two months did the number go up the most?

c) In which months was the number lower than in the month before?

d) What is the difference between the highest and the lowest months?

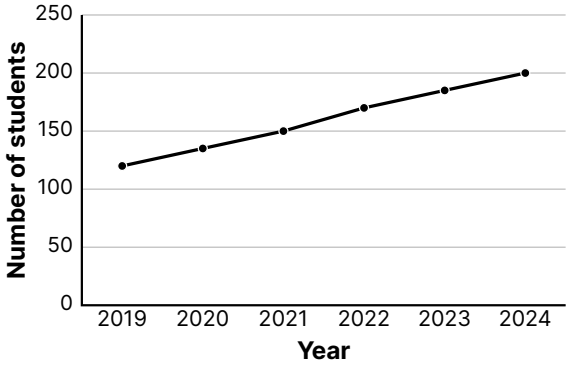
e) How many books were borrowed in the first three months altogether?

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C Spotting a trend ●●●

Remember: A trend is the overall direction of the data over a long time. Small ups and downs do not matter. Ask: is it going up, going down, or staying about the same?

3. The graph shows the number of students at a school each year.



a) Is the number of students increasing, decreasing or staying about the same?

b) How much did the number change from 2019 to 2024?

c) If the trend carries on, about how many students would you expect in 2025? Explain.

4. The table shows the number of children at a chess club each week.

Week	1	2	3	4	5	6
Children	32	30	33	31	32	30

a) Is the number of children increasing, decreasing or staying about the same? Explain how you know.

★ Extension

5. Give one example of something you could measure every day and show on a time-series graph.

6. Why is a time-series graph better than a table for seeing a trend?

7. A graph shows the temperature at midday each day for a week. The line goes down on three days in a row. Describe the trend over those days.

8. On a time-series graph the value is 12 on Monday and 20 on Friday. By how much did it increase?

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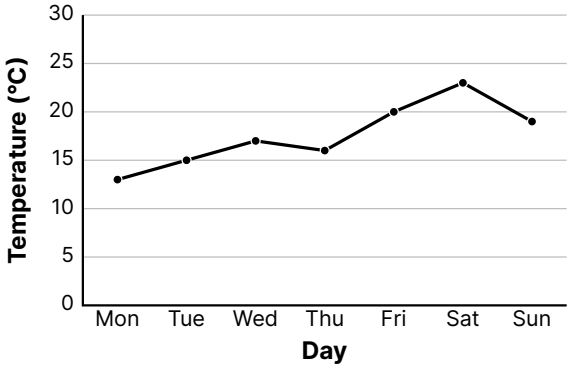
Name _____ **Date** _____ No calculator **Version 2** Scan to practise this topic online

KEY WORDS time-series graph *(kauwhata raupapa wā)* · increase *(pikinga)* · decrease *(hekenga)* · trend *(ia)*

A Reading a time-series graph ●○○

Remember: A time-series graph shows how something changes over time. Time goes along the bottom. Find the time, go up to the point, then read across to the scale.

1. The graph shows the temperature at midday in a town on each day of one week.



a) What was the temperature on Wed? _____

b) What was the temperature on Fri? _____

c) Which day was the warmest? _____

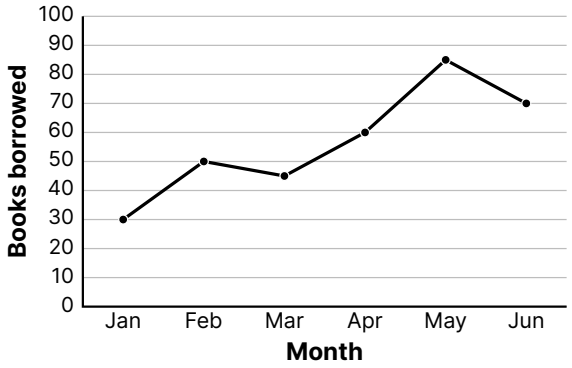
d) Which day was the coldest? _____

e) On which day was the temperature 15°C? _____

B Changes ●●○

Remember: When the line goes up, the amount is increasing. When it goes down, the amount is decreasing. A steeper line means a bigger change.

2. The graph shows the number of books borrowed from a school library each month.



a) How many more books were borrowed in Apr than in Mar?

b) Between which two months did the number go up the most?

c) In which months was the number lower than in the month before?

d) What is the difference between the highest and the lowest months?

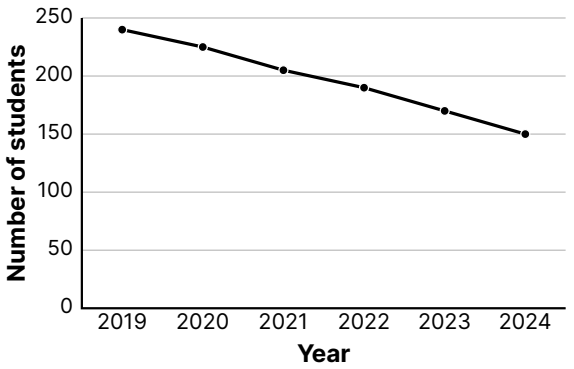
e) How many books were borrowed in the first three months altogether?

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C Spotting a trend ●●●

Remember: A trend is the overall direction of the data over a long time. Small ups and downs do not matter. Ask: is it going up, going down, or staying about the same?

3. The graph shows the number of students at a school each year.



a) Is the number of students increasing, decreasing or staying about the same?

b) How much did the number change from 2019 to 2024?

c) If the trend carries on, about how many students would you expect in 2025? Explain.

4. The table shows the number of children at a chess club each week.

Week	1	2	3	4	5	6
Children	25	27	24	26	25	27

a) Is the number of children increasing, decreasing or staying about the same? Explain how you know.

★ Extension

5. Give one example of something you could measure every day and show on a time-series graph.

6. Why is a time-series graph better than a table for seeing a trend?

7. A graph shows the temperature at midday each day for a week. The line goes down on three days in a row. Describe the trend over those days.

8. On a time-series graph the value is 15 on Monday and 27 on Friday. By how much did it increase?

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